

Types of Discrete Probability Distributions

3.4.1 Bernoulli distribution

DEFINITION 3.10

A random variable with two possible values, 0 and 1, is called a **Bernoulli variable**, its distribution is **Bernoulli distribution**, and any experiment with a *binary outcome* is called a **Bernoulli trial**.

e.g., tossing a coin, winning or losing a game, $\dots$.

If $P(1) = p$ is the probability of a *success*, then $P(0) = q = 1 - p$ is the probability of a *failure*. We can then compute the expectation and variance as

$$\begin{aligned} \mathbf{E}(X) &= \sum_x P(x) = (0)(1 - p) + (1)(p) = p, \\ \text{Var}(X) &= \sum_x (x - p)^2 P(x) = (0 - p)^2(1 - p) + (1 - p)^2 p \\ &= p(1 - p)(p + 1 - p) = p(1 - p). \end{aligned}$$

**Bernoulli
distribution**

$$\begin{aligned} p &= \text{probability of success} \\ P(x) &= \begin{cases} q = 1 - p & \text{if } x = 0 \\ p & \text{if } x = 1 \end{cases} \\ \mathbf{E}(X) &= p \\ \text{Var}(X) &= pq \end{aligned}$$

3.4.2 Binomial distribution

A **binomial probability distribution** results from a **Binomial Experiment** that meets all the following requirements:

- (a) The experiment consists of n identical trials.
- (b) Each trial has exactly two outcomes, which are usually referred to as success (S) or failure (F).
- (c) The probability of success on each individual trial is always the same number $P(S) = p$. (The probability of failure is then always $P(F) = 1 - p$. It is traditional to use the notation $P(F) = q = 1 - p$.)
- (d) The trials are independent.

Then the probability distribution of the binomial random variable X , the number of successes in n independent trials, is

$$P(x) = \mathbf{P}\{X = x\} = \binom{n}{x} p^x q^{n-x}, \quad x = 0, 1, \dots, n, \quad (3.9)$$

Expected Value and Variance of Binomial Distribution:

Any Binomial variable X can be represented as a sum of independent Bernoulli variables,

$$\mathbf{E}(X) = \mathbf{E}(X_1 + \dots + X_n) = \mathbf{E}(X_1) + \dots + \mathbf{E}(X_n) = p + \dots + p = np$$

and

$$\text{Var}(X) = \text{Var}(X_1 + \dots + X_n) = \text{Var}(X_1) + \dots + \text{Var}(X_n) = npq.$$

**Binomial
distribution**

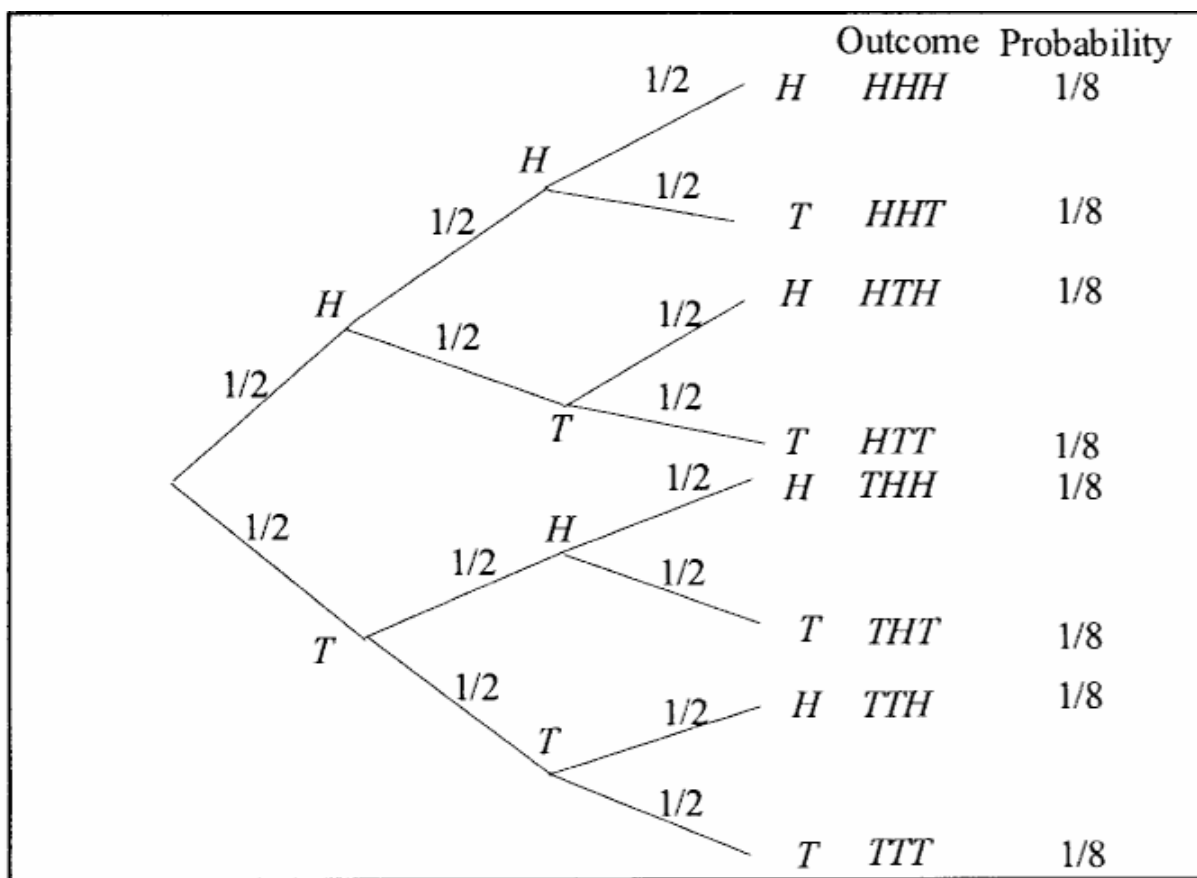
n	=	number of trials
p	=	probability of success
$P(x)$	=	$\binom{n}{x} p^x q^{n-x}$
$\mathbf{E}(X)$	=	np
$\text{Var}(X)$	=	npq

Example: A fair coin is tossed three times and the number of heads $\mathbf{X}$ is recorded. The experiment is a binomial experiment since all of the following hold:

- (a) There are $n = 3$ identical trials (coin tosses).
- (b) Each trial has two outcomes: heads (a success, S) or tails (a failure, F).
- (c) The probability of success is the same on each trial; in this case, $P(S) = P(H) = .50$ for each toss.
- (d) Successive tosses of a fair coin are independent.

Thus X is a binomial random variable. □

Below is the tree for three tosses of a fair coin. Probabilities for each branch are included.



Let X be the number of heads observed.

$$P(X = 2) = (\text{number of branches with 2 heads}) \frac{1}{8} = 3 \left(\frac{1}{8} \right) = \frac{3}{8}. \quad \square$$

Using Binomial Distribution:

$$P(X = 2) = \binom{3}{2} \left(\frac{1}{2} \right)^2 \left(\frac{1}{2} \right)^1 = 3 \left(\frac{1}{8} \right) = \frac{3}{8}$$

Example 3.16. As part of a business strategy, randomly selected 20% of new internet service subscribers receive a special promotion from the provider. A group of 10 neighbors signs for the service. What is the probability that at least 4 of them get a special promotion?

Or

$$P(X \geq 4) = 1 - P(X \leq 3) = 1 - (P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3))$$

$$\text{For } x = 0, 1, 2, 3: P(X = x) = \binom{10}{x} (0.2)^x (0.8)^{10-x}$$

$$P(0) = (0.8)^{10} \approx 0.1074$$

$$P(1) = 10(0.2)(0.8)^9 \approx 0.2684$$

$$P(2) = 45(0.2)^2(0.8)^8 \approx 0.3020$$

$$P(3) = 120(0.2)^3(0.8)^7 \approx 0.2013$$

$$P(X \leq 3) \approx 0.1074 + 0.2684 + 0.3020 + 0.2013 = 0.8791$$

$$P(X \geq 4) = 1 - 0.8791 = 0.1209$$

From WP

Example 5.2: The probability that a patient recovers from a rare blood disease is 0.4. If 15 people are known to have contracted this disease, what is the probability that (a) at least 10 survive, (b) from 3 to 8 survive, and (c) exactly 5 survive?

Solution: Let X be the number of people who survive.

$$\begin{aligned} \text{(a)} \quad P(X \geq 10) &= 1 - P(X < 10) = 1 - \sum_{x=0}^9 b(x; 15, 0.4) = 1 - 0.9662 \\ &= 0.0338 \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad P(3 \leq X \leq 8) &= \sum_{x=3}^8 b(x; 15, 0.4) = \sum_{x=0}^8 b(x; 15, 0.4) - \sum_{x=0}^2 b(x; 15, 0.4) \\ &= 0.9050 - 0.0271 = 0.8779 \end{aligned}$$

$$\begin{aligned} \text{(c)} \quad P(X = 5) &= b(5; 15, 0.4) = \sum_{x=0}^5 b(x; 15, 0.4) - \sum_{x=0}^4 b(x; 15, 0.4) \\ &= 0.4032 - 0.2173 = 0.1859 \end{aligned}$$

a. For $x = 10, 11, 12, 13, 14, 15$

$$P(X = x) = \binom{15}{x} (0.4)^x (0.6)^{15-x}$$

$$P(10) = \binom{15}{10} (0.4)^{10} (0.6)^5$$

$$P(11) = \binom{15}{11} (0.4)^{11} (0.6)^4$$

$$P(12) = \binom{15}{12} (0.4)^{12} (0.6)^3$$

$$P(13) = \binom{15}{13} (0.4)^{13} (0.6)^2$$

$$P(14) = \binom{15}{14} (0.4)^{14} (0.6)^1$$

$$P(15) = \binom{15}{15} (0.4)^{15}$$

$$P(X \geq 10) \approx 0.0338$$

b. For $x = 3, 4, 5, 6, 7, 8$

$$P(X = x) = \binom{15}{x} (0.4)^x (0.6)^{15-x}$$

$$P(3 \leq X \leq 8) = \sum_{x=3}^8 P(X = x)$$

$$P(3 \leq X \leq 8) \approx \boxed{0.8796}$$

Example 5.3: A large chain retailer purchases a certain kind of electronic device from a manufacturer. The manufacturer indicates that the defective rate of the device is 3%.

- (a) The inspector randomly picks 20 items from a shipment. What is the probability that there will be at least one defective item among these 20?
- (b) Suppose that the retailer receives 10 shipments in a month and the inspector randomly tests 20 devices per shipment. What is the probability that there will be exactly 3 shipments each containing at least one defective device among the 20 that are selected and tested from the shipment?

Solution: (a) Denote by X the number of defective devices among the 20. Then X follows a $b(x; 20, 0.03)$ distribution. Hence,

$$\begin{aligned} P(X \geq 1) &= 1 - P(X = 0) = 1 - b(0; 20, 0.03) \\ &= 1 - (0.03)^0 (1 - 0.03)^{20-0} = 0.4562. \end{aligned}$$

(b) Exactly 3 shipments (out of 10) have at least one defective

In this case, each shipment can either contain at least one defective item or not. Hence, testing of each shipment can be viewed as a Bernoulli trial with $p = 0.4562$ from part (a). Assuming independence from shipment to shipment

and denoting by Y the number of shipments containing at least one defective item, Y follows another binomial distribution $b(y; 10, 0.4562)$. Therefore,

$$P(Y = 3) = \binom{10}{3} 0.4562^3 (1 - 0.4562)^7 = 0.1602.$$



Example 5.5: Find the mean and variance of the binomial random variable of Example 5.2, and then use Chebyshev's theorem (on page 137) to interpret the interval $\mu \pm 2\sigma$.

Solution: Since Example 5.2 was a binomial experiment with $n = 15$ and $p = 0.4$, by Theorem 5.1, we have

$$\mu = (15)(0.4) = 6 \text{ and } \sigma^2 = (15)(0.4)(0.6) = 3.6.$$

Taking the square root of 3.6, we find that $\sigma = 1.897$. Hence, the required interval is $6 \pm (2)(1.897)$, or from 2.206 to 9.794. Chebyshev's theorem states that the number of recoveries among 15 patients who contracted the disease has a probability of at least $3/4$ of falling between 2.206 and 9.794 or, because the data are discrete, between 2 and 10 inclusive. J

Theorem 4.10: **(Chebyshev's Theorem)** The probability that any random variable X will assume a value within k standard deviations of the mean is at least $1 - 1/k^2$. That is,

$$P(\mu - k\sigma < X < \mu + k\sigma) \geq 1 - \frac{1}{k^2}.$$

For $k = 2$, the theorem states that the random variable X has a probability of at least $1 - 1/2^2 = 3/4$ of falling within two standard deviations of the mean. That is, three-fourths or more of the observations of any distribution lie in the interval $\mu \pm 2\sigma$. Similarly, the theorem says that at least eight-ninths of the observations of any distribution fall in the interval $\mu \pm 3\sigma$.

3.4.5 Poisson distribution

DEFINITION 3.14

The number of rare events occurring within a fixed period of time has **Poisson distribution**.

Note: The Poisson distribution arose from a mathematical formulation called the Poisson Process, published by Siméon-Denis Poisson in 1837.

Poisson Process: The Poisson process counts the number of events:

1. occurring in a fixed time or space,
2. when events occur independently and,
3. at a constant average rate.

**Poisson
distribution**

$$\begin{aligned} \lambda &= \text{frequency, average number of events} \\ P(x) &= e^{-\lambda} \frac{\lambda^x}{x!}, \quad x = 0, 1, 2, \dots \\ \mathbf{E}(X) &= \lambda \\ \text{Var}(X) &= \lambda \end{aligned}$$

Example 5.18: Ten is the average number of oil tankers arriving each day at a certain port. The facilities at the port can handle at most 15 tankers per day. What is the probability that on a given day tankers have to be turned away?

Solution: Let X be the number of tankers arriving each day. Then, using Table A.2, we have

$$P(X > 15) = 1 - P(X \leq 15) = 1 - \sum_{x=0}^{15} p(x; 10) = 1 - 0.9513 = 0.0487.$$

